INCH-POUND

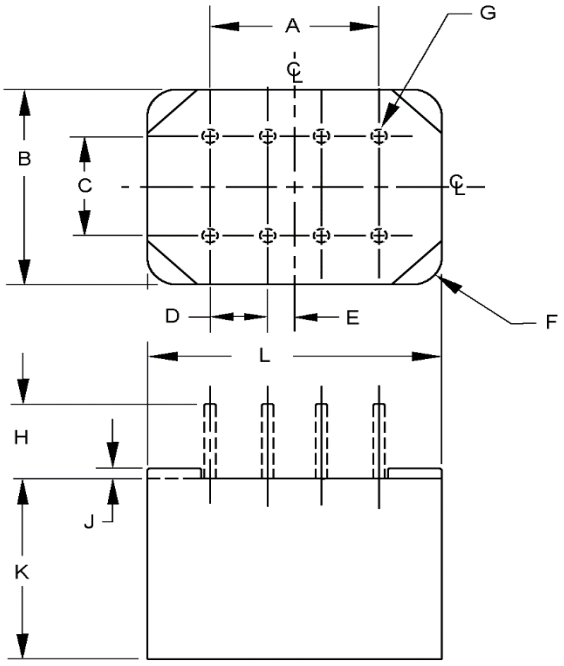
MIL-PRF-27/167E
w/AMENDMENT 2
20 June 2024
SUPERSEDING
MIL-PRF-27/167E
w/AMENDMENT 1
1 November 2016

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, AUDIO FREQUENCY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-27.



Dim	Inches	mm
A	.600 ±.010	15.24 ±.25
B	.750 Max.	19.05 Max.
C	.400 ±.010	10.16 ±.25
D	.200 Typ.	5.08 Typ.
E	.100 ±.010	2.54 ±.25
F	.084 R ±.015	2.13 ±.38
H	.281 ±.031	7.14 ±.79
J	.031 ±.015	.79 ±.38
K	.563 Max.	14.30 Max.
L	.875 Max.	22.22 Max.

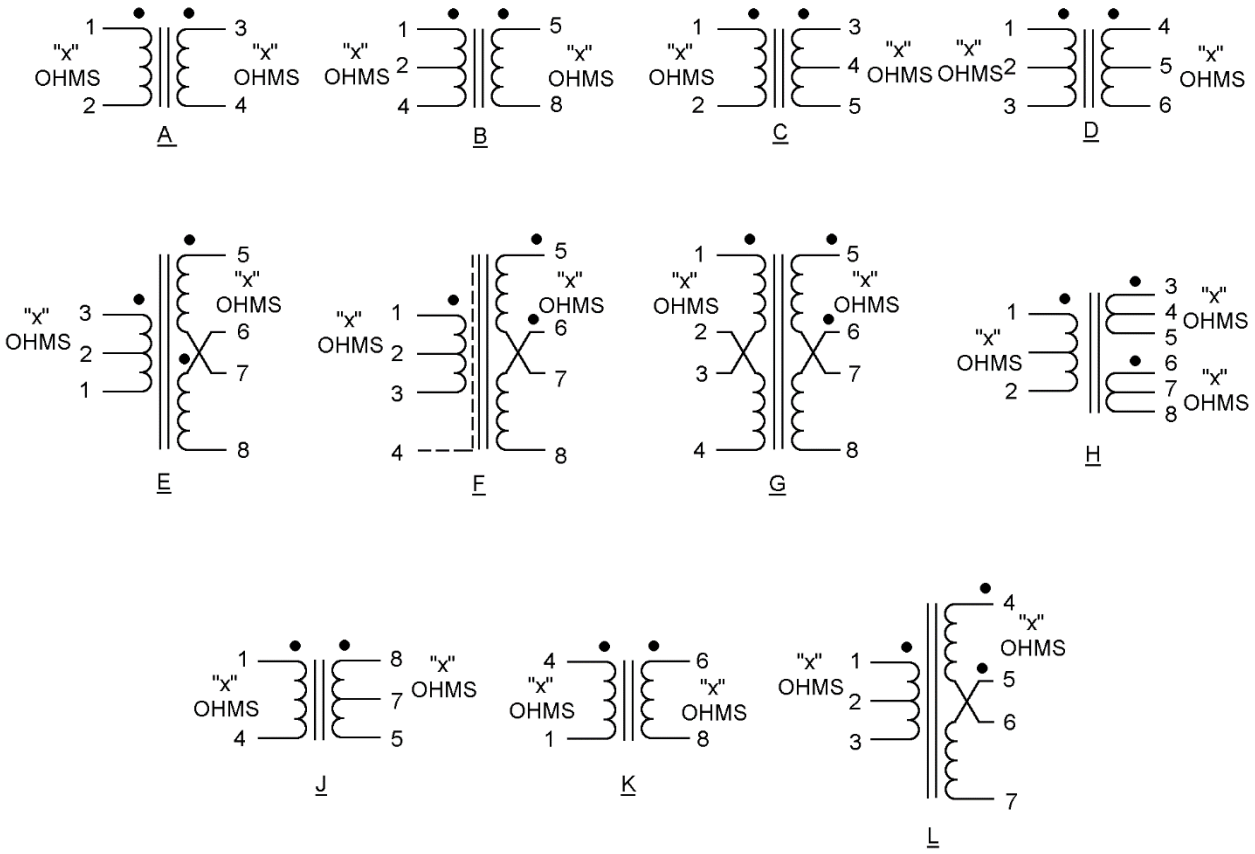
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side(s) and/or top of case.
4. Primary and secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations.

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

FSC 5950

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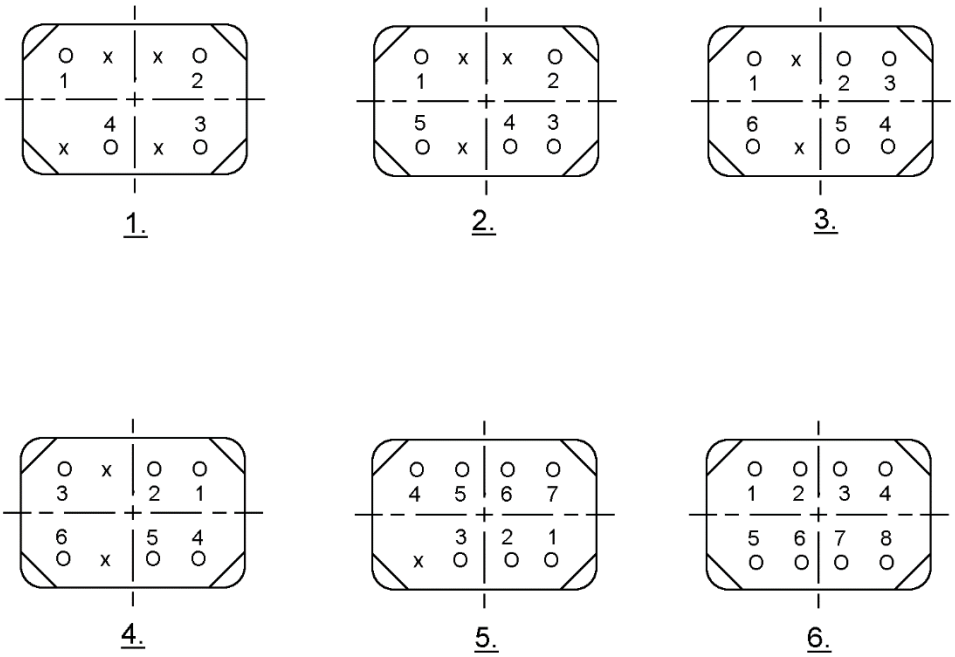


WORKING VOLTAGE (PEAK): 175 V
ALTITUDE: 75,000 FT MAX
(SEE TABLE I FOR VALUE OF “X”)

CIRCUIT DIAGRAM AND MARKING

FIGURE 1. Dimensions and configurations - Continued.

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NOTE: Pins not used are removed. These are indicated by "X".

FIGURE 2. Pin arrangement.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Working voltage (peak): 175 volts peak.

Frequency range: 300 hertz to 20 kilohertz.

Design and construction:

Dimensions and configurations: See figure 1.

Duty cycle: Continuous.

Case: Plastic.

Terminals: Pin type, .040 ±.005 inch diameter.

Material: Type LW(---)C5N in accordance with MIL-STD-1276.

Weight: 20 grams maximum.

Altitude: 75,000 feet maximum.

Operating temperature range: -55°C to +105°C.

Terminal strength: MIL-STD-202-211, test condition A, 2.0 pounds.

Vibration (high frequency): MIL-STD-202-204, test condition D.

Dielectric withstanding voltage:

At sea level: 500 V rms.

At reduced barometric pressure: 300 V rms.

Electrical characteristics: See table II.

Frequency response: See table II.

Reference frequency: 1 kilohertz.

Insertion loss: At the rated power level at 1 kilohertz, the insertion loss shall be as specified in table II.

No load (centertap voltage unbalance only): With 1 volt, 5 kilohertz across the primary the voltage unbalance shall be ±1 percent.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/167-(dash number from table I).

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TABLE I. Electrical ratings.

Dash No.	1/ 2/ Primary Impedance (ohms)	1/ 2/ Secondary Impedance (ohms)	Primary DCR ±25% (ohms)	Secondary DCR ±25% (ohms)	Max Power Level (mW)	Primary dc current (mA)	Pin arrangement (see fig. 2)	Circuit Diagram (see fig. 1)
01	10 k 25 k	200 500	2,500	34	100	3 1.5	1	A
02	50 k 200	62.5 k 250 k	13.5	3,600	5	0	6	K
03	100 Split	100 Split	5.9	6.9	100	10	6	G
04	500 Split 600 Split	12.8 Split 16 Split	36.2	1.5	100	8	6	G
05	500 600	50 60	50	5	100	3.5	6	K
06	500 Split 600 Split	50 Split 60 Split	36	6.5	100	3.5	6	G
07	500 CT	600 CT	26	70	100	10	6	D
08	500 Split	600 Split	30	42	100	10	6	G
09 3/	600 CT	600 Split	42	29	100	10	6	F
10	600 Split	2 k Split	40	113	100	5	6	G
11	600	600 CT and 600 CT	50	50 and 50	25	5	6	H
12 3/	600 Split	8 k CT	55	484	50	3.2	6	F
13	800 Split 1,200 Split	20 k Split 30 k Split	110	800	50	3	6	G
14	1,000 1,200	50 60	5	90	100	6	6	K
15	1,000 Split 1,200 Split	200 Split 240 Split	102	19	100	3	6	G
16	1,200 1,500	3.2 4.0	70	0.45	100	7	6	K
17	1.5 k CT	600 CT	70	65	100	2.5	3	D
18	4 k CT	600 Split	155	47	100	2.6	5	L
19	4 k Split 5 k Split	12.8 Split 16 Split	327	1.5	100	2	6	G
20	10 k	3.2	800	0.65	100	2	6	K
21	8 k Split 10 k Split	12.8 Split 16 Split	600	1.5	100	2	6	G

See footnotes at end of table.

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TABLE I. Electrical ratings - Continued.

Dash No.	1/ 2/ Primary Impedance (ohms)	1/ 2/ Secondary Impedance (ohms)	Primary DCR ±25% (ohms)	Secondary DCR ±25% (ohms)	Max Power Level (mW)	Primary dc current (mA)	Pin arrangement (see fig. 2)	Circuit Diagram (see fig. 1)
22	10 k	16	800	2.7	100	4	6	K
23	10 k CT 12 k CT	3.2 4.0	770	0.73	100	3 1.5	6	B
24	10 k Split 25 k Split	200 Split 500 Split	2,500	34	100	2 1	6	G
25	10 k CT 25 k CT	200 CT 500 CT	560	22	100	2	3	D
26	10 k 12 k	500 CT 600 CT	800	95	100	4	6	J
27	10 k Split 12 k Split	500 Split 600 Split	800	95	100	4	6	G
28	10 k CT	2 k CT	1,200	45	100	1	2	C
29	10 k CT 12 k CT	1,500 CT 1,800 CT	800	300	100	4	4	D
30	10 k CT 12 k CT	10 k CT 12 k CT	560	650	100	1	5	L
31	10 k Split 12 k Split	10 k Split 12 k Split	560	650	100	1	6	G
32	10 k	90 k	710	3,150	32	0.25	6	K
33	20 k 30 k	800 1,200	800	110	100	0.5	6	K
34	20 k CT 30 k CT	800 CT 1,200 CT	800	110	100	1	3	D
35	30 k	50	2,875	4.6	100	1	6	K
36	40 k CT 50 k CT	400 Split 500 Split	1,900	43	100	0.5	6	E
37	40 k Split 50 k Split	400 Split 500 Split	1,900	43	100	0.5	6	G
38	100 k	60	3,500	3.3	100	0.5	6	K
39 3/	100 k CT 200 k CT	100 Split 200 Split	3,290	4.1	5	0	6	F
40	200 K	1 k	4,000	190	5	0	6	K
41	200 K CT	1 k CT	4,000	200	5	0	3	D
42 3/	200 k CT	1 k Split	4,000	200	5	0	6	F

1/ Impedance values written one above the other indicates a range of matching impedances over which the parts will give satisfactory performance as long as the impedance ratio is maintained.
2/ Where windings are listed as split, one-fourth of the listed impedance is available by paralleling the winding.
3/ Electrostatic shield, shield ratio 5:1 minimum at 20 kHz.
4/ Qualification testing and approval to M27/167-01 shall be sufficient to grant qualification approval to M27/167-01 through M27/167-42, inclusive.

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TABLE II. Electrical characteristics.

Dash no.	Frequency response at 300 Hz to 20 kHz ±3 dB			Resonance, second resonant frequency (min)	Insertion loss at 1 kHz (max)	Polarity Same on terminals
	Z_S (ohms)	Z_L (ohms)	E_L (volts)			
01	25 K	500	7.07	20 kHz	3.5	1, 3
02	50	62.5 K	17.7	20 kHz	4.0	4, 6
03	100	100	3.16	50 kHz	3.5	1, 5, 6
04	600	16	1.26	50 kHz	3.5	1, 5, 6
05	600	60	2.45	50 kHz	3.5	4, 6
06	600	60	2.45	50 kHz	3.5	1, 5, 6
07	500	600	7.75	50 kHz	3.0	1, 4
08	500	600	7.75	50 kHz	3.5	1, 5, 6
09	600	600	7.75	50 kHz	3.5	1, 5, 6
10	600	2,000	14.1	50 kHz	3.5	1, 5, 6
11	600 (1-2)	1.2 k (3-5) 1.2 k (6-8)	3.87 3.87	50 kHz	3.5	1, 3, 6
12	600	8,000	20	20 kHz	3.5	1, 5, 6
13	1,200	30 k	38.7	20 kHz	3.5	1, 5, 6
14	1,200	60	2.45	20 kHz	3.5	4, 6
15	1,200	30 k	38.7	20 kHz	3.5	1, 5, 6
16	1,200	4.0	0.63	20 kHz	4.0	4, 6
17	1,500	600	7.75	20 kHz	3.5	1, 4
18	1,500	600	7.75	20 kHz	3.5	1, 4, 5
19	5 k	16	1.26	20 kHz	3.5	1, 5, 6
20	10 k	3.2	0.56	20 kHz	4.0	4, 6
21	10 k	16	1.26	20 kHz	3.5	1, 5, 6
22	10 k	16	1.26	20 kHz	4.0	4, 6
23	12 k	4	0.63	20 kHz	5.0	1, 5
24	25 k	500	7.07	20 kHz	3.5	1, 5, 6
25	25 k	500	7.07	20 kHz	3.0	1, 4
26	12 k	600	7.75	20 kHz	3.5	1, 8
27	12 k	600	7.75	20 kHz	3.5	1, 5, 6
28	10 k	2 k	14.1	20 kHz	3.5	1, 3
29	12 k	1.8 k	13.4	20 kHz	3.5	1, 4
30	12 k	12 k	34.6	20 kHz	3.5	1, 4, 5
31	12 k	12 k	34.6	20 kHz	3.5	1, 5, 6
32	10 k	90 k	52	20 kHz	3.5	4, 6
33	30 k	1.2 k	11	20 kHz	3.5	4, 6
34	30 k	1.2 k	11	20 kHz	3.5	1, 4
35	30 k	50	2.23	20 kHz	3.5	4, 6
36	50 k	500	7.07	20 kHz	3.5	3, 5, 6
37	50 k	500	7.07	20 kHz	3.5	1, 5, 6
38	100 k	60	2.45	20 kHz	3.5	4, 6
39	200 k	200	1.0	20 kHz	3.0	1, 5, 6
40	200 k	1 k	2.23	20 kHz	3.5	4, 6
41	200 k	1 k	2.23	20 kHz	3.5	1, 4
42	200 k	1 k	2.23	20 kHz	3.5	1, 5, 6

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Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202-204 MIL-STD-202-211 MIL-STD-1276

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA – CC

(Project 5950-2024-018)

Review activities:
Army – AR, MD, MI
Navy – AS, MC, OS, SH
Air Force – 19

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